

Test Case ID	Feature	Input	Expected Output	Actual Output	Result	Evidence / Link
TC-01	Login & Authentication Valid login (Admin)	Username: osama25, password: osama123	API returns 200 OK with JWT; App navigates to Admin Panel	API returns 200 OK with JWT; App navigates to Admin Panel	Pass	[]
TC-02	Login & Authentication Valid login (Manager)	Username: Anas11, password: anas123	API returns 200 OK with JWT; App navigates to Manager Panel	API returns 200 OK with JWT; App navigates to Manager Panel	Pass	[]
TC-03	Login & Authentication Valid login (Farmer)	Username: Noor28, password: Noor123	API returns 200 OK with JWT; App navigates to Farmer Panel	API returns 200 OK with JWT; App navigates to Farmer Panel	Pass	[]
TC-04	Login & Authentication Non-Existent User	Username: "UnknownUser", Password: "AnyPassword"	API Returns 401Unauthorized("InvalidUsername or password"), stay at login screen	API Returns 401Unauthorized("InvalidUsername or password"), stay at login screen	Pass	[]
TC-05	Login & Authentication Empty Credentials	Username""', password""'	App prevents request or API returns 400 Bad Request; Input validation highlights empty fields.	App prevents request or API returns 400 Bad Request; Input validation highlights empty fields.	pass	[]
TC-06	Logout Storage & token cleanup	Click Logout button	JWT is purged from Secure Storage and backend marks DeviceToken as inactive/null	JWT is purged from Secure Storage and backend marks DeviceToken	Pass	[]

			in the DB to stop notifications.	as inactive/null in the DB to stop notifications.		
TC-07	Add Sensor Data Valid Data submission	JSON: { Temp: 24, Humidity: 60, SoilMoisture: 55, N: 40, P: 35, K: 50, pH: 6.5 }	API returns 201 Created; Record visible in DB with current timestamp.			
TC-08	Add Sensor Data Valid Invalid PlantId	JSON: { Temp: 24, Humidity: 60, SoilMoisture: 55, N: 40, P: 35, K: 50, pH: 6.5 }	API returns 404 Not Found or 400 Bad Request due to foreign key constraint.			
TC-09	Add Sensor Data (Hardware Integration)	Real-time transmission from ESP32 via Wi-Fi	API successfully parses raw sensor string and logs data to the database			
TC-10	Add Sensor (Data without API Key)	JSON: { Temp: 24, Humidity: 60, SoilMoisture: 55, N: 40, P: 35, K: 50, pH: 6.5 }	API Returns 401 Unauthorized			
TC-11	Add Sensor Data (Out of Range)	Ph: 100, ...	API Returns 400 bad request			
TC-12	Assigned Plants For Farmer	Log in as "Farmer_A" (Farmer)	App displays only plants assigned to Farmer_A;			

			Plants assigned to Farmer_B are hidden.			
TC-12	Assigned Plants For Farmer (No assignments)	Login as a new Farmer with 0 assignments	"No plants assigned"			
TC-13	All Greenhouse Plants for Manager	Login as "Manager_X" choose a greenhouse and navigates to all plants	App displays all plants located within Greenhouse_X,			
TC-14	All Greenhouse Plants for Manager (No Plants)	Login as "Manager_X" choose a greenhouse and navigates to all plants	App displays "no plants for this greenhouse"			
TC-15	Generate Report (PDF)	Date Range: "Last 7 Days", Plants: tomato, Metric: "Soil Moisture"	API generates a PDF; File downloads successfully to mobile device storage.			
TC-16	Generate Report (Excel)	Date Range: "Last 7 Days", Plants: tomato, Metric: "Soil Moisture"	API generates a Excel; File downloads successfully to mobile device storage.			
TC-17.	Generate Report Empty Data Range	Date Range: Future Dates (e.g., 2027)	API Generates a file with no data			

TC-18	Generate Report (Multi-Metric Support)					
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